Data Aggregation:

World Happiness Dataset

Presented by Zaw Min Tun

Given Dataset

World Happiness Dataset:

	Country	Happiness_Score	GDP_per_Capita	Social_Support	Healthy_Life_Expectancy	Freedom_to_Make_Choices	Generosity	Perceptions_of_Corruption
1	Norway	6.25	1.39	0.82	81.6	0.69	0.01	0.71
2	Denmark	3.61	1.27	0.43	66.9	0.48	0.43	0.53
3	Iceland	4.68	0.87	0.54	63.4	0.71	0.41	0.72
4	Switzerland	4.46	0.67	0.57	68.8	0.93	0.32	0.52
5	Finland	6.67	1.55	0.45	75.4	0.58	0.16	0.10
6	Netherlands	6.41	0.87	0.54	72.6	0.45	0.38	0.36
7	Canada	7.34	0.60	0.46	49.6	1.00	0.07	0.12
8	New Zealand	3.87	0.61	0.57	41.3	0.66	0.26	0.84
9	Australia	5.31	1.43	0.78	53.2	0.36	0.27	0.80
10	Sweden	3.63	1.16	0.62	51.2	0.33	0.57	0.77
11	United States	4.44	1.39	0.62	48.9	0.38	0.53	0.35
12	United Kingdom	5.67	1.30	0.53	79.6	0.74	0.16	0.15
13	Germany	3.61	1.09	0.56	76.8	0.85	0.30	0.80
14	France	4.36	1.57	0.96	53.2	0.60	0.11	0.86
15	Japan	6.29	0.92	0.79	67.5	0.34	0.55	0.17
16	South Korea	5.84	1.11	0.77	56.6	0.57	0.52	0.49
17	India	4.45	1.41	0.50	78.4	1.00	0.18	0.16
18	China	6.03	1.18	0.84	59.3	0.67	0.38	0.71
19	Brazil	6.98	1.45	0.50	51.1	0.98	0.37	0.71
20	South Africa	3.53	1.14	0.63	50.4	0.90	0.09	0.20

GDP Categories

```
WITH categorised AS (
    SELECT Country, Happiness_Score, GDP_per_Capita,
        CASE
            WHEN GDP_per_Capita < 0.9  THEN 'Low'
            WHEN GDP_per_Capita < 1.3  THEN 'Medium'
            ELSE   'High'
        END AS GDP_Category
    FROM read_csv_auto('{DATA}')),


    with_avg AS (
        SELECT Country, Happiness_Score,
            GDP_per_Capita, GDP_Category,
            ROUND(AVG(Happiness_Score)
                OVER (PARTITION BY GDP_Category), 2)
                AS Avg_Happiness_Category,
                RANK() OVER (PARTITION BY GDP_Category
                ORDER BY Happiness_Score DESC)
                AS Rank_in_Same_GDP_Category
         FROM categorised)

    SELECT *
    FROM with_avg
    ORDER BY
        CASE GDP_Category
        WHEN 'High' THEN 1
        WHEN 'Medium' THEN 2
        ELSE 3 END,
        Rank_in_Same_GDP_Category
```

First Common Table Expression (CTE) ‘categorised’. This creates categories to refer — categorised by GDP_per_Capita.

Just loading the data as table (DuckDB)

Calculating average happiness score for each group defined in ‘categorised’, and ranking each record by Happiness_Score in each group.

Final query referring the ‘with_avg’ CTE defined above, ordering ‘High’ category of GDP_per_Capita first, followed by ‘Medium’ and then ‘Low’

GDP Categories

Country	Happiness_Score	GDP_per_Capita	GDP_Category	Avg_Happiness_Category	Rank_in_Same_GDP_Category
Brazil	6.98	1.45	High	5.52	1
Finland	6.67	1.55	High	5.52	2
Norway	6.25	1.39	High	5.52	3
United Kingdom	5.67	1.3	High	5.52	4
Australia	5.31	1.43	High	5.52	5
India	4.45	1.41	High	5.52	6
United States	4.44	1.39	High	5.52	7
France	4.36	1.57	High	5.52	8
Japan	6.29	0.92	Medium	4.65	1
China	6.03	1.18	Medium	4.65	2
South Korea	5.84	1.11	Medium	4.65	3
Sweden	3.63	1.16	Medium	4.65	4
Denmark	3.61	1.27	Medium	4.65	5
Germany	3.61	1.09	Medium	4.65	5
South Africa	3.53	1.14	Medium	4.65	7
Canada	7.34	0.6	Low	5.35	1
Netherlands	6.41	0.87	Low	5.35	2
Iceland	4.68	0.87	Low	5.35	3
Switzerland	4.46	0.67	Low	5.35	4
New Zealand	3.87	0.61	Low	5.35	5

High GDP Category: GDP_per_Capita >= 1.3
Medium GDP Category: 0.9 <= GDP_per_Capita < 1.3
Low GDP Category: GDP_per_Capita < 0.9

Corruption Perception

```
SELECT
        Corruption_Group,
        COUNT(*) AS Country_Count,
        ROUND(AVG(Happiness_Score), 2) AS Avg_Happiness,
        ROUND(AVG(GDP_per_Capita), 2) AS Avg_GDP,
        ROUND(AVG(Social_Support), 2) AS Avg_Social_Support,
        ROUND(AVG(Healthy_Life_Expectancy), 2) AS Avg_Life_Expectancy,
        ROUND(AVG(Freedom_to_Make_Choices), 2) AS Avg_Freedom,
        ROUND(AVG(Generosity), 2) AS Avg_Generosity
FROM (
        SELECT
            Country,
            Happiness_Score,
            GDP_per_Capita,
            Social_Support,
            Healthy_Life_Expectancy,
            Freedom_to_Make_Choices,
            Generosity,
            Perceptions_of_Corruption,
            CASE WHEN Perceptions_of_Corruption >=
            (
                SELECT AVG(Perceptions_of_Corruption)
                FROM read_csv_auto('{DATA}')
            )
            THEN 'High Corruption Perception'
            ELSE 'Low Corruption Perception'
            END AS Corruption_Group
        FROM read_csv_auto('{DATA}')
    ) GROUP BY Corruption_Group
    ORDER BY Corruption_Group
```

Main query showing two groups — High Corruption Perception and Low Corruption Perception — counting the countries and calculating the average for each feature

Subquery to define ‘Corruption Perception’ groups

Sub-subquery to calculate average Perceptions_of_Corruption

Corruption Perception

--- Query 2: High vs Low Corruption — Average Comparisons ---

Corruption_Group	Country_Count	Avg_Happiness	Avg_GDP	Avg_Social_Support	Avg_Life_Expectancy	Avg_Freedom	Avg_Generosity
High Corruption Perception	11	4.8	1.15	0.65	60.62	0.66	0.31
Low Corruption Perception	9	5.63	1.14	0.59	64.33	0.66	0.29

Corruption Perception Details

```
SELECT
    Corruption_Group,
    Country,
    Happiness_Score,
    GDP_per_Capita,
    Perceptions_of_Corruption,
    RANK() OVER (PARTITION BY Corruption_Group
    ORDER BY Happiness_Score DESC) AS Rank_in_Group
FROM (
        SELECT
            Country,
            Happiness_Score,
            GDP_per_Capita,
            Perceptions_of_Corruption,
            CASE WHEN Perceptions_of_Corruption >=
            (
                SELECT AVG(Perceptions_of_Corruption)
                FROM read_csv_auto('{DATA}')
            )
            THEN 'High Corruption Perception'
            ELSE 'Low Corruption Perception'
            END AS Corruption_Group
        FROM read_csv_auto('{DATA}'))
        ORDER BY Corruption_Group, Rank_in_Group
```

Main query showing the data categorising High Corruption Perception or Low Corruption Perception and ranking the data for each group

Subquery to define ‘Corruption Perception’ groups

Sub-subquery to calculate average Perceptions_of_Corruption

Corruption Perception Details

--- Query 3: Each Country Sorted within Corruption Group ---

Corruption_Group	Country	Happiness_Score	GDP_per_Capita	Perceptions_of_Corruption	Rank_in_Group
High Corruption Perception	Brazil	6.98	1.45	0.71	1
High Corruption Perception	Norway	6.25	1.39	0.71	2
High Corruption Perception	China	6.03	1.18	0.71	3
High Corruption Perception	Australia	5.31	1.43	0.8	4
High Corruption Perception	Iceland	4.68	0.87	0.72	5
High Corruption Perception	Switzerland	4.46	0.67	0.52	6
High Corruption Perception	France	4.36	1.57	0.86	7
High Corruption Perception	New Zealand	3.87	0.61	0.84	8
High Corruption Perception	Sweden	3.63	1.16	0.77	9
High Corruption Perception	Denmark	3.61	1.27	0.53	10
High Corruption Perception	Germany	3.61	1.09	0.8	10
Low Corruption Perception	Canada	7.34	0.6	0.12	1
Low Corruption Perception	Finland	6.67	1.55	0.1	2
Low Corruption Perception	Netherlands	6.41	0.87	0.36	3
Low Corruption Perception	Japan	6.29	0.92	0.17	4
Low Corruption Perception	South Korea	5.84	1.11	0.49	5
Low Corruption Perception	United Kingdom	5.67	1.3	0.15	6
Low Corruption Perception	India	4.45	1.41	0.16	7
Low Corruption Perception	United States	4.44	1.39	0.35	8
Low Corruption Perception	South Africa	3.53	1.14	0.2	9

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Thank you !!!

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